SALARY VISUALIZATION

Data-Driven Insights for Income Patterns and Socio-Economic Impact

SECTOR
Social Impact / Socio-Economic Analytics

TEAM ID
G3 | **SECTION:** C

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EXECUTIVE SUMMARY

PROBLEM
Income inequality is influenced by multiple socio-economic factors. Understanding these patterns helps identify high-income drivers and groups needing support for income mobility.

APPROACH
The project cleans and analyzes Adult Census data, computes key KPIs, and builds an interactive Tableau dashboard to visualize income patterns across demographic, occupational, and financial dimensions.

KEY INSIGHTS
- Education strongly increases the likelihood of earning > $50K.
- Occupation and capital gains are major income drivers.
- Gender and marital status show visible income gaps.

KEY RECOMMENDATIONS
- Invest in education and upskilling.
- Promote high-skill career development.
- Encourage financial literacy and investments.
- Support inclusion and fair pay initiatives.

IMPACT & VALUE

INCOME MOBILITY
Education & upskilling increase the share of high-income individuals.

WORKFORCE DEVELOPMENT
Training for high-paying occupations enables better income groups.

INEQUALITY REDUCTION
Gender & marital income gaps highlight unequal earning patterns.

FINANCIAL GROWTH
Capital gains strongly increase high-income probability.

POLICY EFFICIENCY
Income patterns help identify groups needing targeted support.

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3. Sector and Business Context

Sector Overview

The project focuses on socio-economic analysis using census data to understand income distribution patterns.

Stakeholders

Policy makers, workforce planners, and economic analysts.

Importance

Understanding income drivers enables:

- Better policy design
- Targeted interventions
- Efficient workforce planning

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4. Problem Statement and Objectives

Problem Statement

Identify key socio-economic factors influencing income levels and build a system to predict whether an individual earns above $50K.

Objectives

- Analyze demographic and employment features
- Identify key income drivers
- Build predictive models
- Generate actionable insights

Scope

Limited to Adult Census dataset.

Success Criteria

- Strong analytical insights
- Effective predictive model
- Decision-oriented dashboard

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5. Data Description

Source

Adult Census Dataset

Dataset Overview

- ~15,000+ records
- 15 features

Key Features

- Age, Education, Occupation
- Workclass, Marital Status
- Capital Gain/Loss
- Hours per Week
- Income (target variable)

Data Issues

- Missing values (“?”)
- Categorical inconsistencies
- Class imbalance

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6. Cleaning and Transformation

This section details the ETL (Extract, Transform, Load) pipeline and data engineering process used to prepare the dataset for analysis and modeling.

6.1 Objective

The ETL (Extract, Transform, Load) pipeline was designed to convert raw Adult Census data into a clean, consistent, and analysis-ready dataset. The primary goal was to ensure data reliability, reproducibility, and seamless integration with downstream analytics and visualization workflows.

6.2 Pipeline Architecture

The pipeline follows a structured, modular flow:

Raw Data → Extraction → Cleaning → Transformation → Validation → Processed Dataset

Each stage is isolated to ensure maintainability, ease of debugging, and scalability.

6.3 Extraction Layer

- Source: `data/raw/adult.csv`
- Implemented controlled ingestion to securely load raw data into the processing environment
- Performed initial schema checks to validate column structure and data consistency before transformation
- Ensured dataset integrity by verifying row counts and feature alignment

6.4 Transformation and Data Cleaning

6.4.1 Schema Standardization

- Converted all column names to snake_case format (e.g., marital-status → marital_status)
- Ensures compatibility with Python pipelines, SQL systems, and downstream tools

6.4.2 Missing Value Handling

- Identified missing values represented as `'?'`
- Replaced missing values using mode-based imputation per column
- Preserved natural distribution of categorical variables
- Prevented propagation of NaN values to avoid downstream inconsistencies

6.4.3 Data Cleaning

- Removed leading and trailing whitespaces using vectorized operations
- Eliminated duplicate records to reduce noise and prevent skewed results

6.4.4 Feature Engineering

- **Target Encoding:**
  Converted income categories (`>50K`, `<=50K`) into binary format (1, 0) for classification
- **Age Bucketing:**
  Transformed continuous age values into categorical groups:
  Youth, YoungAdult, Adult, Senior
  using conditional binning (`pd.cut()`)

This improved interpretability and supported segmentation analysis.

6.5 Automation and Pipeline Engineering

- Transitioned from exploratory Jupyter notebooks to a production-ready Python script: `etl_pipeline.py`
- Implemented modular and reusable code structure
- Integrated logging mechanisms for monitoring and debugging
- Enabled execution via command-line interface:

python scripts/etl_pipeline.py

This transition ensures portability from experimental workflows to scalable and repeatable data processing.

6.6 Data Quality Assurance

The pipeline enforces strict quality guarantees:

- No missing values remain in the dataset
- Consistent schema and naming conventions
- Duplicate-free records
- Clean and standardized categorical values

6.7 Output Dataset

- Location: `data/processed/adult_cleaned.csv`
- Fully processed and analysis-ready dataset

Optimized for:

- Exploratory Data Analysis (EDA)
- Machine Learning models
- Tableau dashboard visualization

6.8 Key Engineering Decisions

- **Dropping `fnlwgt`:**
  Removed due to low analytical relevance
- **Mode Imputation:**
  Preserves categorical integrity
- **Binary Target Encoding:**
  Simplifies classification tasks
- **Age Bucketing:**
  Reduces noise and improves interpretability
- **Notebook → Script Transition:**
  Ensures reproducibility and production-readiness

6.9 Pipeline Impact

The ETL pipeline establishes a robust data foundation that enables accurate analysis, reliable insights, and scalable integration with visualization tools. It ensures that all downstream components operate on a clean and consistent dataset, significantly improving overall project quality.

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7. KPI Framework

The KPI (Key Performance Indicator) framework was designed to measure income distribution and identify key drivers influencing high-income individuals (>50K). These KPIs form the foundation of both analysis and dashboard decision-making.

7.1 Core KPI: High Income Rate

Definition:
Percentage of individuals earning more than $50K.

Formula:

```
SUM(
 IF [Income Label] = ">50K" THEN 1 ELSE 0 END
)
/ COUNT([Income Label])
```

Purpose:
This is the primary KPI used across the dashboard to evaluate income distribution and compare different demographic and socio-economic groups.

7.2 Gender Income Gap

Definition:
Difference in high-income rate between male and female individuals.

Purpose:
Measures income disparity across genders and helps identify inequality patterns in earnings.

7.3 Top Occupation (High Income)

Definition:
Occupation group with the highest proportion of individuals earning >50K.

Purpose:
Identifies high-paying job categories and helps understand structural income differences across professions.

7.4 Top Education Level (High Income)

Definition:
Education category with the highest percentage of individuals earning >50K.

Purpose:
Highlights the impact of education on income and supports insights related to skill development and economic mobility.

7.5 Marital Status Income Gap

Definition:
Difference in income distribution across different marital status categories.

Purpose:
Analyzes how social and demographic factors influence earning potential.

7.6 Supporting Metrics

Number of Records

COUNT([Income Label])

Used to represent total population size and ensure proportional comparisons.

Income Distribution (Binary Classification)

- <=50K
- 50K

Used as the base segmentation for all KPIs and visualizations.

7.7 KPI Importance

These KPIs collectively help:

- Quantify income distribution across the dataset
- Identify key socio-economic drivers of high income
- Compare performance across demographic groups
- Support data-driven decision-making in workforce planning

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8. Exploratory Analysis

8.1 Objective

The objective of the Exploratory Data Analysis (EDA) was to understand the structure, distribution, and relationships within the dataset, and to identify key factors influencing income classification (<=50K vs >50K). This step forms the foundation for subsequent statistical analysis and predictive modeling.

8.2 Data Overview

The dataset contains demographic and socio-economic attributes of individuals such as:

- Age
- Workclass
- Education
- Occupation
- Marital Status
- Hours per Week
- Income Label (Target Variable)

8.2.1 Initial Inspection

The dataset was examined using:

- `.head()` → to preview records
- `.info()` → to check data types and null values
- `.describe()` → to understand numerical distributions

8.2.2 Data Cleaning

The following preprocessing steps were performed:

Handling Missing Values

- Missing values in categorical columns (e.g., Workclass, Occupation) were identified
- These were either:
    - Removed
    - Or treated as a separate category (depending on approach)

Data Type Corrections

- Ensured categorical variables were treated correctly
- Converted target variable into usable format

Duplicate Handling

- Checked and removed duplicate rows if present

8.3 Univariate Analysis

8.3.1 Numerical Features

Age

- Distribution analyzed using histogram
- Observed:
    - Majority of individuals fall between 25–50 years
    - Right-skewed distribution

Hours per Week

- Most individuals work between 35–45 hours
- Few extreme values observed (outliers)

8.3.2 Categorical Features

Income Distribution

- Highly imbalanced dataset:
    - Majority: <=50K
    - Minority: >50K

Education

- Most common categories:
    - HS-grad
    - Some-college
    - Bachelors

Workclass

- Dominated by:
    - Private sector

Marital Status

- Categories like:
    - Married-civ-spouse
    - Never-married

Occupation

- Wide range of job types
- Highly granular categories

8.4 Bivariate Analysis

8.4.1 Income vs Age

- Older individuals tend to have higher income probability
- Peak income observed in middle-age groups

8.4.2 Income vs Education

- Strong positive relationship observed
- Higher education levels correspond to higher income

8.4.3 Income vs Occupation

- Managerial and professional roles:
    - Higher proportion of >50K
- Manual and service roles:
    - Mostly <=50K

8.4.4 Income vs Gender

- Males have a higher proportion of high-income individuals
- Gender disparity observed

8.4.5 Income vs Hours per Week

- Slight positive relationship
- Higher earners tend to work more hours

8.5 Feature Engineering

To improve interpretability, new features were created:

8.5.1 Age Group

- IF age < 25 → Youth
- IF age < 40 → YoungAdult
- IF age < 60 → Adult
- ELSE → Senior

8.5.2 Education Group

Grouped into:

- Low Education
- School Level
- College Level
- Higher Education

8.5.3 Occupation Group

Grouped into:

- High Skill Jobs
- Skilled Technical
- Service & Sales
- Public Sector
- Manual Labor

8.6 Outlier Analysis

- Extreme values observed in:
    - Hours per week
    - Age
- These were:
    - Either retained (if realistic)
    - Or noted as potential anomalies

8.7 Key Patterns Identified

- Education is one of the strongest predictors of income
- Occupation type significantly influences earning potential
- Gender disparity exists in high-income distribution
- Income peaks in middle age
- Work hours have moderate impact on income

8.8 Data Limitations

- Class imbalance in income variable
- Presence of categorical noise (many small groups)
- Potential bias in demographic features

8.9 Conclusion

The EDA provided critical insights into the dataset, highlighting key relationships between demographic factors and income. These findings guided feature engineering, statistical analysis, and the development of predictive models in subsequent stages.

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9. Statistical Analysis

To move beyond descriptive insights, statistical and machine learning methods were applied.

9.1 Predictive Modeling using XGBoost

An XGBoost classifier was developed to predict whether an individual earns more than $50K annually based on demographic, educational, and occupational features.

XGBoost was chosen due to its high performance and ability to capture complex relationships between variables.

Model Performance:

- Accuracy: ~85%
- Balanced performance across both income classes

Threshold Optimization:
Instead of using the default probability threshold (0.5), multiple thresholds were evaluated.

A threshold of **0.4** was selected as optimal because:

- It improves recall for high-income individuals (>50K)
- Maintains acceptable overall accuracy

From a business perspective, identifying potential high-income individuals is more valuable than minimizing false positives.

9.2 Feature Importance Analysis

The model identified the most influential factors affecting income:

- Relationship status is the strongest predictor
- Capital gain strongly correlates with higher income
- Education level significantly impacts earning potential
- Marital status indicates stability and dual-income effects

These findings validate and strengthen EDA insights.

9.3 Customer Segmentation using Clustering

K-Means clustering was applied to group individuals into meaningful segments.

Three distinct segments were identified:

- **Standard Workforce**
  Majority population with moderate income
- **Emerging High Earners**
  Individuals with increasing income potential
- **Elite High-Income Group**
  High-income individuals with strong financial characteristics

This segmentation enables targeted decision-making.

9.4 Cluster Visualization using PCA

PCA was used to visualize clusters.

Initially, distortion occurred due to large variance in features such as capital gain. After applying feature scaling, clearer cluster separation was observed.

This confirms distinct socio-economic segments.

9.5 What-if Analysis

A what-if analysis was conducted to understand feature impact.

Example:

- Increasing education level increases probability of earning >50K

This improves interpretability and supports decision-making.

9.6 Business Value

Machine learning enhances the dataset from descriptive to actionable.

- Predicts high-income individuals
- Identifies high-potential segments
- Supports workforce planning
- Enables data-driven decisions

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10. Dashboard Walkthrough

10.1 Dashboard 1: Income Analysis

10.1.1 Objective

The Tableau dashboard was developed to analyze income distribution (≤50K vs >50K) based on demographic and socio-economic factors such as age, gender, education, occupation, and workclass. The goal was to derive interpretable insights and present them through an interactive and visually structured dashboard.

10.1.2 Data Preparation in Tableau

The dataset imported into Tableau was preprocessed and included derived categorical fields for better visualization.

10.1.2.1 Created / Used Fields

1. Income Label
Used as the primary binary classification:

- <=50K
- 50K

2. Age Group (Derived)
Categorized age into interpretable segments:

```
IF [age] < 25 THEN "Youth"

ELSEIF [age] < 40 THEN "YoungAdult"

ELSEIF [age] < 60 THEN "Adult"

ELSE "Senior"

END
```

3. Education Group (Derived)
Grouped detailed education levels into meaningful categories:

```
IF [Education] IN ("Preschool", "1st-4th", "5th-6th", "7th-8th")

THEN "Low Education"


ELSEIF [Education] IN ("9th", "10th", "11th", "12th", "HS-grad")

THEN "School Level"


ELSEIF [Education] IN ("Some-college", "Assoc-acdm", "Assoc-voc")

THEN "College Level"


ELSE "Higher Education"

END
```

4. Occupation Group (Derived)
Grouped occupations for clarity:

```
IF [Occupation] IN ("Exec-managerial", "Prof-specialty")

THEN "High Skill Jobs"


ELSEIF [Occupation] IN ("Tech-support", "Craft-repair")

THEN "Skilled Technical"


ELSEIF [Occupation] IN ("Sales", "Other-service", "Handlers-cleaners")

THEN "Service & Sales"


ELSEIF [Occupation] IN ("Protective-serv", "Adm-clerical")

THEN "Public Sector"


ELSE "Manual Labor"

END
```

5. Number of Records (Calculated Field)

```
COUNT([Income Label])
```

6. High Income Rate (Key Metric)

```
SUM(

  IF [Income Label] = ">50K" THEN 1 ELSE 0 END

)

/
```

COUNT([Income Label])

Formatted as percentage.

10.1.3 KPIs Implemented

Displayed at the top of the dashboard:

1. High Income Rate
   Percentage of individuals earning >50K
2. Gender Income Gap
   Difference in high income rate between male and female
3. Top Occupation (High Income)
   Occupation group with highest income rate
4. Top Education (High Income)
   Education group with highest income rate
5. Marital Status Income Gap
   Difference in income rate across marital categories

10.1.4 Charts Implemented

10.1.4.1 Income Rate by Education

- Type: Horizontal Bar Chart
- Measure: High Income Rate
- Dimension: Education Group
- Insight: Income probability increases significantly with education level

10.1.4.2 Income Rate by Occupation

- Type: Bar Chart
- Dimension: Occupation Group
- Measure: High Income Rate
- Insight: High skill jobs have significantly higher income probability compared to manual and service roles

10.1.4.3 Workclass Treemap

- Type: Treemap
- Dimension: Workclass
- Measure: High Income Rate
- Insight: Self-employed and government roles show varying income patterns

10.1.4.4 Gender-wise Income Distribution

- Type: Pie Chart
- Dimension: Sex
- Color: Income Label
- Measure: Percentage of Total
- Insight: Males have a higher proportion of >50K income compared to females

10.1.4.5 Age Distribution vs Income

- Type: Histogram / Bar
- Dimension: Age (bin) or Age Group
- Measure: Count
- Insight: Income tends to peak in middle age groups

10.1.4.6 Average Hours vs Income

- Type: Bar Chart
- Measure: AVG(hours_per_week)
- Split by: Income Label
- Insight: Higher earners tend to work slightly more hours

10.1.5 Filters Implemented

1. Income Level (Single Value Buttons)
2. Gender (Single Value Buttons)
3. Age Group (Dropdown)
4. Education Group (Multiple Values Dropdown)
5. Marital Status (Dropdown)

All filters applied globally using:
Apply to Worksheets → All Using This Data Source

10.1.6 Dashboard Design

Layout Structure

- Title
- KPI Row
- Main Charts
- Secondary Charts

Design Elements

- Background: Dark grey
- Accent Color: Muted red (#C05A49)
- KPI cards: Light background
- Consistent color coding

10.1.7 Interactivity Features

- Global filter application
- Dynamic chart updates
- Centralized filter bar

10.1.8 Key Insights

- Income strongly correlates with education level
- High-skill occupations have higher income probability
- Gender income disparity exists
- Middle-aged individuals dominate high income
- Work hours moderately influence income

10.1.9 Conclusion

The dashboard effectively presents income distribution patterns through structured visualizations and interactivity, enabling intuitive exploration and data-driven insights.

10.2 Dashboard 2: Wealth Signals – Capital Gains & Income

10.2.1 Objective

To analyze how capital gains and capital losses influence income levels and determine whether financial asset behavior is a strong indicator of high earnings.

10.2.2 Key Insights

- Capital gains significantly increase probability of earning >50K (~41% lift)
- Only ~8.7% individuals have capital gains but dominate high-income group
- Capital loss shows minimal influence
- Education amplifies the effect of capital gains
- Absence of capital gains strongly correlates with lower income

10.2.3 Conclusion

Capital gains act as a strong financial indicator of high income, while capital loss has minimal predictive impact. Education and asset ownership together strengthen income outcomes.

10.3 Dashboard 3: Who Becomes a High Earner? – Prediction Insights

10.3.1 Objective

To identify behavioral and demographic drivers of high income probability using model predictions.

10.3.2 Key Insights

- Income probability increases with working hours (peaks ~90 hours/week)
- Self-employed (incorporated) category shows highest income probability (~53.7%)
- High-skill occupations outperform service and labor roles
- Higher education strongly correlates with income
- Private jobs dominate volume but not income probability

10.3.3 Conclusion

High income is driven by a combination of:

- Higher education
- High-skill occupations
- Entrepreneurial workclass
- Longer working hours

10.4 Final Combined Takeaway

The analysis across all dashboards shows that financial assets (capital gains), education, and occupational structure are the strongest predictors of high income. Behavioral factors such as working hours further enhance income probability. High earnings result from interconnected socio-economic and professional factors.

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11. Key Insights

1. Education strongly drives income
2. Capital gain is a major indicator
3. Occupation significantly impacts earnings
4. Gender disparity exists
5. Income peaks in middle age
6. Work hours moderately influence income
7. Relationship status is a key factor
8. Income is multi-factor dependent

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12. Recommendations

Based on the combined insights from exploratory analysis, statistical modeling, and dashboard findings, the following recommendations are proposed:

12.1 Strengthen Access to Higher Education

Education emerged as one of the strongest predictors of income across all analyses. Individuals with higher education levels consistently show a significantly higher probability of earning above $50K.

Recommendation:
Promote access to higher education and professional qualifications through scholarships, policy incentives, and skill development programs.

12.2 Focus on High-Skill and Professional Job Development

Occupation-based analysis revealed that roles such as managerial, professional, and technical jobs have the highest income probabilities.

Recommendation:
Encourage workforce transition towards high-skill roles through targeted training programs, certifications, and industry-aligned skill development initiatives.

12.3 Encourage Financial Asset Creation (Capital Gains)

Dashboard analysis showed that individuals with capital gains have a significantly higher likelihood of being in the high-income group.

Recommendation:
Promote financial literacy, investment awareness, and asset-building strategies to enable individuals to generate wealth beyond primary income sources.

12.4 Support Entrepreneurship and Self-Employment

Self-employed individuals, especially in incorporated categories, demonstrate higher income probabilities compared to traditional employment roles.

Recommendation:
Encourage entrepreneurship through startup support programs, easier access to credit, and policy incentives for small businesses.

12.5 Target Workforce Upskilling in Low-Income Segments

Segmentation analysis identified a large “Standard Workforce” group with moderate income and limited capital growth.

Recommendation:
Design targeted upskilling programs for this segment to improve earning potential and enable upward mobility into higher-income groups.

13. Limitations and Next Steps

13.1 Limitations

Class Imbalance in Target Variable

The dataset is heavily skewed towards individuals earning <=50K, which may affect model performance and bias predictions toward the majority class.

Limited Feature Representation

The dataset lacks important real-world variables such as:

- Geographic location
- Industry trends
- Economic conditions

This restricts the model’s ability to capture broader socio-economic influences.

Static and Historical Dataset

The dataset represents a fixed snapshot in time and does not account for:

- Changing economic conditions
- Job market evolution
- Inflation and policy changes

High Cardinality in Categorical Variables

Features like occupation and workclass contain many categories, which may introduce noise and reduce model generalization.

13.2 Future Work

Integration of Real-Time and External Data

Enhance analysis by incorporating:

- Economic indicators
- Regional data
- Industry-specific trends

Advanced Modeling Techniques

Explore more advanced approaches such as:

- Ensemble optimization
- Deep learning models
- Hyperparameter tuning

Improved Feature Engineering

Develop more meaningful features such as:

- Experience level
- Skill categories
- Income progression indicators

Fairness and Bias Analysis

Conduct bias detection to evaluate whether predictions are influenced by sensitive attributes such as gender or demographic factors.

Deployment and Decision Systems

Extend the project into a real-world application by:

- Deploying the model as an API
- Integrating with dashboards for real-time predictions
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14. Contribution Matrix

(To be filled by team based on GitHub contributions)

Team Member	Dataset and Sourcing	ETL and Cleaning	EDA and Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT and Viva
Atharva Tiwari							Owner
Rachit Gupta						Owner	
Agrima Gusain			Owner		Owner		
Tanmay Singh	Owner						
R Ashrith		Owner					
Ashish Kushwaha				Owner			